Recoil transport and environmental exposure of black-hole merger remnants

Venkata Hemanth Athota[1,*]

[1]*Independent researcher*

Anisotropic gravitational-wave emission can impart a recoil velocity to the black hole formed by a merger. In a stellar cluster, the resulting motion depends on the merger location, the progenitor's center-of-mass velocity, the cluster's mass and velocity distributions, and gravitational scattering. Remaining bound does not determine when a remnant enters a dense region or how long it occupies that region. We formulate a two-stage description of this problem. A merger-conditioned phase-space transport model combines smooth gravity, external differential acceleration, mean collisional drift, and stochastic velocity diffusion. Finite-region entry, residence, and species-resolved environmental exposure are then defined on the resulting trajectories, with explicit treatment of termination, censoring, and host uncertainty. The formulation specifies the physical inputs and trajectory information needed for subsequent numerical investigation without selecting a cluster model or claiming calibrated transport predictions. Its immediate objective is to determine how recoil and the stellar environment jointly control the remnant's time-dependent occupation of dense cluster regions.

I. INTRODUCTION

A black hole formed by a merger can receive a recoil from anisotropic gravitational-wave emission. Within a stellar cluster, its subsequent motion reflects both that recoil and the state of the system in which the merger occurs. The progenitor's center-of-mass motion, merger position, host potential, and gravitational scattering determine the remnant's evolving position and velocity. A gravitationally bound remnant need not remain near the center, and entry into a dense region does not imply that the remnant comes to rest there.

The distinction matters whenever the surrounding population influences subsequent evolution. Residence measures the time spent in a specified region, whereas density exposure weights the trajectory by the material encountered along it. Neither is fixed by a retention flag alone. Their calculation requires a dynamical description with enough information to distinguish repeated excursions, continued wandering, and changes in the environment during the remnant's motion.

The necessary kinetic ingredients are established. Rosenbluth collision integrals relate velocity drift and diffusion to a field-particle distribution [1]; their gravitational application permits anisotropic velocity distributions [2]. Direct calculations of kicked black holes illustrate limitations of simplified dynamical-friction descriptions and show that individual encounters can increase a remnant's energy [3]. These results motivate treating mean drift and fluctuations together, while testing the regime in which a local diffusion closure is adequate. They do not supply a calibrated prescription for an arbitrary cluster.

We formulate two linked problems. Step 1 defines merger-conditioned transport in position and velocity, with an explicit matter inventory, reference frame, collision closure, and termination rule. Step 2 defines finite-region entry, residence, and species-resolved environmental exposure on the resulting histories. Their common output is a timed description of the remnant's occupation of its environment. The immediate question is which launch and host information must be retained to predict these observables at a declared accuracy.

The paper specifies a theoretical calculation rather than reporting evaluated trajectories or fits. No host family, numerical parameters, data-conditioned posterior, or accuracy threshold is selected. Distribution-function families can help supply consistent host models [4], but naming such a family does not determine its parameters, hidden mass components, or collision coefficients. The formulation distinguishes these physical inputs from the observables that a subsequent calculation would measure.

II. STEP 1: RECOIL TRANSPORT

A. Matter distributions and reference frame

Use physical units and cluster-relative coordinates in translating, nonrotating axes. Let $\mathbf{R}(t)$ be the reference position in an inertial frame, and let $(\mathbf{r}, \mathbf{v})$ denote the remnant's relative position and velocity. The host description Θ contains its matter distributions, gravitational boundary conditions, and any supplied temporal evolution.

For species j with individual mass m_j, define a number phase-space distribution $F_j(\mathbf{r}, \mathbf{u}, t \mid \Theta)$. Thus $F_j\, d^3r\, d^3u$ is an expected object count. Its dimensions are inverse volume times inverse velocity cubed. The number and mass densities are

$$n_j(\mathbf{r}, t) = \int F_j(\mathbf{r}, \mathbf{u}, t)\, d^3u, \tag{1}$$

$$\rho_j(\mathbf{r}, t) = m_j n_j(\mathbf{r}, t), \qquad \rho_c = \sum_j \rho_j. \tag{2}$$

The cluster potential satisfies

$$\nabla^2 \Phi_c = 4\pi G \rho_c, \tag{3}$$

together with specified boundary conditions. The tracked remnant's self-field is excluded. An unresolved binary can be represented by its center of mass only on encounter scales supporting that approximation; its mass must not also enter through separately counted components. A continuous mass

* bharadwaj.hemanth.002@gmail.com

spectrum requires a mass integral or a declared binning convention.

The same distributions determine streaming and velocity dispersion:

$$\overline{\mathbf{u}}_j = \frac{1}{n_j}\int \mathbf{u} F_j\, d^3u, \tag{4}$$

$$S_{j,ik} = \frac{1}{n_j}\int (u_i - \overline{u}_{j,i})(u_k - \overline{u}_{j,k}) F_j\, d^3u. \tag{5}$$

These moments are defined where $n_j > 0$. Distinct species may have different concentrations, streaming motions, and anisotropies. Density and a scalar dispersion do not generally determine the full velocity distribution needed for a collision calculation.

For an external gravitational potential $\Phi_{\rm ext}$, the relative external acceleration is

$$\mathbf{a}_{\rm ext,rel} = -\nabla\Phi_{\rm ext}(\mathbf{R}+\mathbf{r},t) - \ddot{\mathbf{R}}(t). \tag{6}$$

If the reference orbit follows $\ddot{\mathbf{R}} = -\nabla\Phi_{\rm ext}(\mathbf{R},t)$, Eq. (6) is the differential tidal acceleration. There are no Coriolis terms in these nonrotating axes. Internal stellar rotation remains possible through F_j.

This is a test-remnant description in a supplied background. Poisson's equation does not evolve the host. A frozen host needs a validity interval; an evolving host needs mutually consistent distributions and potential at each epoch. Backreaction, collective response, and energy feedback require additional equations and are not obtained by introducing a time argument alone.

B. Merger-conditioned initial conditions

Set elapsed time zero at the remnant-producing merger. The launch conditions are

$$\mathbf{r}(0) = \mathbf{r}_{\rm m}, \qquad \mathbf{v}(0) = \mathbf{v}_{\rm COM} + \mathbf{v}_{\rm k}, \tag{7}$$

$$m_{\rm r}(0) = m_{\rm rem}. \tag{8}$$

Here $\mathbf{v}_{\rm COM}$ is the progenitor binary's center-of-mass velocity, not its internal orbital velocity. The recoil vector $\mathbf{v}_{\rm k}$ must be transformed into the cluster frame before addition. Equation (7) uses nonrelativistic cluster kinematics; a relativistic merger calculation supplies recoil and remnant mass.

Let $\mathcal{D}$ denote observational or simulation constraints and $\mathcal{M}$ the merger-selection rule and relevant assembly information. The initial law is joint:

$$z_0 = (\mathbf{r}_{\rm m}, \mathbf{v}_{\rm COM}, \mathbf{v}_{\rm k}, m_{\rm rem}), \tag{9}$$

$$z_0 \sim \pi_0(z_0 \mid \Theta, \mathcal{D}, \mathcal{M}). \tag{10}$$

No independence among these variables or from the host is imposed. In particular, the initial specific angular momentum $\mathbf{r}_{\rm m}\times(\mathbf{v}_{\rm COM}+\mathbf{v}_{\rm k})$ need not vanish. A noncentral trajectory need not pass through the exact center.

A retained-only population is meaningful only after specifying the escape criterion and the interval over which retention is assessed. Initial negative cluster-relative energy does not by itself establish permanent retention in an evolving tidal environment. The base transport process is defined before any selection on future survival. Statistics conditioned on that survival require conditioning or reweighting its full path law; changing the launch distribution alone while leaving subsequent stochastic evolution unchanged is insufficient.

C. Velocity drift and diffusion

Define the smooth acceleration $\mathbf{a}_g = -\nabla\Phi_{\rm c} + \mathbf{a}_{\rm ext,rel}$. A local diffusion description takes the Itô form

$$d\mathbf{r} = \mathbf{v}\, dt, \tag{11}$$

$$d\mathbf{v} = (\mathbf{a}_g + \mathbf{A})\, dt + \mathbf{B}\, d\mathbf{W}, \tag{12}$$

$$\mathbf{D} = \mathbf{B}\mathbf{B}^{\mathsf{T}}. \tag{13}$$

The three Wiener components satisfy $\mathbb{E}[dW_i\, dW_k] = \delta_{ik}\, dt$. After subtracting smooth gravity, the first and second collisional increments define

$$A_i = \lim_{\Delta t\to 0} \frac{\mathbb{E}[\Delta v_i^{\rm coll} \mid \mathbf{r},\mathbf{v},t]}{\Delta t}, \tag{14}$$

$$D_{ik} = \lim_{\Delta t\to 0} \frac{\mathbb{E}[\Delta v_i^{\rm coll}\Delta v_k^{\rm coll} \mid \mathbf{r},\mathbf{v},t]}{\Delta t}. \tag{15}$$

These limits belong to a coarse-grained approximation; they do not resolve the duration of each gravitational encounter. The dimensions of $\mathbf{A}$ and $\mathbf{D}$ are acceleration and velocity squared per time, respectively. Once $\mathbf{A}$ is the complete Itô first moment, an additional noise-induced drift must not be appended.

Between discrete transitions, remnant mass is fixed on each trajectory. At fixed $\mathcal{Q} = (m_{\rm r}, \Theta, \mathcal{D}, \mathcal{M})$, let $p(\mathbf{r},\mathbf{v},t \mid \mathcal{Q})$ denote its phase-space density. The interior evolution is

$$\begin{aligned}\partial_t p = &-\partial_{r_i}(v_i p) - \partial_{v_i}[(a_{g,i} + A_i)p] \\ &+ \frac{1}{2}\partial_{v_i}\partial_{v_k}(D_{ik}p).\end{aligned} \tag{16}$$

Repeated Cartesian indices are summed. The factor of one half follows from Eq. (13). Initial, boundary, and termination conditions are needed to define a solution. When launch masses vary, each mass-dependent operator must retain its own conditioning; replacing it by an operator evaluated at a mean mass is not generally equivalent.

D. A candidate weak-scattering closure

For local, uncorrelated weak deflections, a leading-logarithm Rosenbluth closure supplies both coefficients from the same matter distribution [1, 2]. At fixed position and time, define

$$h_j(\mathbf{v}) = \int \frac{F_j(\mathbf{r},\mathbf{u},t)}{|\mathbf{v}-\mathbf{u}|}\, d^3u, \tag{17}$$

$$g_j(\mathbf{v}) = \int F_j(\mathbf{r},\mathbf{u},t)|\mathbf{v}-\mathbf{u}|\, d^3u. \tag{18}$$

For the number normalization in Eq. (1),

$$A_i = 4\pi G^2 \sum_j m_j(m_r + m_j) \ln\Lambda_j \, \partial_{v_i} h_j, \tag{19}$$

$$D_{ik} = 4\pi G^2 \sum_j m_j^2 \ln\Lambda_j \, \partial_{v_i}\partial_{v_k} g_j. \tag{20}$$

A mass-normalized distribution instead absorbs a factor of m_j into each Rosenbluth potential. Mixing those conventions would change the collision coefficients.

The Coulomb logarithm $\ln\Lambda_j = \ln(b_{\max,j}/b_{\min,j})$ represents a declared range of encounter impact parameters. It is held fixed during the velocity integrals and differentiations above. Arbitrary velocity-dependent cutoffs require a consistent derivation rather than an unqualified substitution. Neither cutoff values nor a prescription are selected here.

Writing $\mathbf{w} = \mathbf{v} - \mathbf{u}$, the derivative kernels are

$$\partial_{v_i} h_j = -\int F_j \frac{w_i}{w^3}\, d^3u, \tag{21}$$

$$\partial_{v_i}\partial_{v_k} g_j = \int F_j \left(\frac{\delta_{ik}}{w} - \frac{w_i w_k}{w^3}\right) d^3u. \tag{22}$$

For nonnegative distributions and positive Coulomb logarithms, $\mathbf{D}$ is symmetric and positive semidefinite. Streaming and anisotropy enter through F_j; a Maxwellian is not imposed. Mean collisional drift need not oppose the cluster-frame velocity in every anisotropic or streaming population.

The closure excludes strong encounters and collective or long-range correlations. Its applicability depends on the scales and populations contributing to relaxation. Kicked-black-hole calculations showing departures from simplified friction models [3] therefore motivate independent validation, not an assumption that adding these coefficients is sufficient.

E. Energy, termination, and probability accounting

For fixed remnant mass, define specific cluster-relative energy

$$\epsilon = \frac{1}{2}|\mathbf{v}|^2 + \Phi_c(\mathbf{r}, t). \tag{23}$$

Itô's rule gives

$$d\epsilon = \left[\partial_t \Phi_c + \mathbf{v}\cdot\mathbf{a}_{\text{ext,rel}} + \mathbf{v}\cdot\mathbf{A} + \frac{1}{2}\operatorname{tr}\mathbf{D}\right] dt + \mathbf{v}^{\mathsf{T}}\mathbf{B}\, d\mathbf{W}. \tag{24}$$

The diffusion contribution is required. Individual trajectories can gain energy, and this energy is not a conserved tidal escape criterion. No condition resets position or velocity upon core entry.

For a common physical horizon T, define the modeled endpoint

$$T_* = \min\{T, \tau_{\text{esc}}, \tau_{\text{strong}}, \tau_{\text{valid}}\}. \tag{25}$$

The stopping times represent declared escape, a strong encounter requiring an unavailable transition model, and loss of validity of the supplied description. A strong-encounter stop needs a detector, resolved objects, or a justified event model; missing encounter information is not evidence of no encounters. Weak scattering and explicitly resolved interactions must partition deflections without double-counting.

Let $p_{\text{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q)$ include only paths with $t < T_*$. Then

$$\int p_{\text{alive}}\, d^3r\, d^3v = S(t \mid Q) = \Pr(t < T_* \mid Q). \tag{26}$$

Survival here means continued inclusion in the transport description. It does not mean that the black hole ceases to exist when a path is stopped. The transport operator requires the corresponding absorbing boundary or justified killing mechanism. A history-dependent stop requires an augmented state or a justified history-dependent treatment; it does not automatically close Eq. (16) in position and velocity.

The density in Eq. (26) is conditional on fixed remnant mass. A joint mass density is obtained by multiplication with the initial mass marginal π_m:

$$p^{\text{joint}}_{\text{alive}}(\mathbf{r}, \mathbf{v}, m, t \mid \Theta, \mathcal{D}, \mathcal{M}) = \pi_m(m \mid \Theta, \mathcal{D}, \mathcal{M}) p_{\text{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q_m), \tag{27}$$

where $Q_m = (m, \Theta, \mathcal{D}, \mathcal{M})$. Integrating a fixed-mass conditional density over mass without this weight would not represent the launch ensemble.

III. STEP 2: RESIDENCE AND ENVIRONMENTAL EXPOSURE

A. Finite-region entry and re-entry

Specify an operational region by a level-set function,

$$C(t) = \{\mathbf{r} : g(\mathbf{r}, t) < 0\}. \tag{28}$$

The function is negative inside, positive outside, and zero on the boundary. This g describes geometry and is distinct from the velocity-space potentials g_j in Eq. (18). A sphere may be represented by $g = |\mathbf{r}| - r_c(t)$, but neither a shape nor a radius is selected. A projected observational core radius is not automatically the required spatial boundary.

For an initially interior launch, define first exit and subsequent re-entry as

$$\tau_{\text{out}} = \inf\{t > 0 : g(\mathbf{r}_t, t) > 0\}, \tag{29}$$

$$\tau_{\text{re}} = \inf\{t > \tau_{\text{out}} : g(\mathbf{r}_t, t) < 0\}. \tag{30}$$

The strict opposite-side conditions exclude an isolated tangential contact. For a continuous path, the infimum identifies the boundary time associated with a sign-changing crossing. An observed re-entry delay is $\tau_{\text{re}} - \tau_{\text{out}}$.

For an initially exterior launch, define first entry by

$$\tau_{\text{ent}} = \inf\{t > 0 : g(\mathbf{r}_t, t) < 0\}. \tag{31}$$

A boundary launch requires a declared initial-side convention. Classification may use the first definite side reached, while recording the contact separately. It must not create an artificial return event. A path that never exits has no observed re-entry, and the convention $\inf \varnothing = +\infty$ does not establish that a censored event is physically impossible.

For a differentiable boundary,

$$\frac{d}{dt} g(\mathbf{r}_t, t) = \partial_t g + \mathbf{v}_t \cdot \nabla g. \tag{32}$$

Boundary evolution can change membership without inward remnant motion. Both the crossing and its origin must remain identifiable. Sign changes remain the definition even when the derivative vanishes. Separate inner and outer event surfaces can control numerical recrossings only if their effects are reported; they do not silently redefine $C(t)$.

Only events observed before the modeled endpoint are recorded as completed. Simultaneous crossing and stopping events require an ordering convention. Return flags and termination causes are separate fields: a path can re-enter the region and later escape.

B. Path-integrated observables

The residence accumulated before termination is

$$C_T = \int_0^{T_*} \mathbf{1}_{\{\mathbf{r}_t \in C(t)\}}\, dt. \tag{33}$$

It has units of time and obeys $0 \le C_T \le T_* \le T$. The ratio C_T/T uses a common horizon across realizations. If a description stops before T because subsequent motion is unresolved, this ratio is not a complete-horizon physical residence fraction. The alternative C_T/T_*, for $T_* > 0$, describes the modeled interval and answers a different question.

For each species, define mass-density exposure

$$J_{j,T} = \int_0^{T_*} \rho_j(\mathbf{r}_t, t)\, dt. \tag{34}$$

Its dimensions are mass times inverse volume times time. A normalized exposure additionally requires a declared positive reference:

$$I_{j,T} = \int_0^{T_*} \frac{\rho_j(\mathbf{r}_t, t)}{\rho_{\mathrm{ref},j}(t)}\, dt, \tag{35}$$
$$\rho_{\mathrm{ref},j}(t) > 0.$$

It has units of time. The reference is a convention, not an automatically inferred central density. Unless it bounds the sampled density, $I_{j,T}/T$ need not lie between zero and one.

Density may vary even while a trajectory remains inside $C(t)$; density outside that region may also be nonzero. Region-restricted exposure can be defined by inserting the same indicator into Eq. (34), but it is a distinct observable. None of these integrals is an encounter count.

C. Time-augmented occupation

Preserve the chronological relation between motion and environment through

$$\mu_T(\mathcal{A}) = \int_0^{T_*} \mathbf{1}_{\{(t,\mathbf{r}_t,\mathbf{v}_t)\in\mathcal{A}\}}\, dt, \tag{36}$$

where $\mathcal{A}$ is a measurable subset of $[0,T] \times \mathbb{R}^3 \times \mathbb{R}^3$. The measure has total mass T_* and units of time. Equations (33) and (34) are weighted integrals of that same measure:

$$C_T = \int \mathbf{1}_{\{g(\mathbf{r},t)<0\}}\, \mu_T(dt, d^3r, d^3v), \tag{37}$$

$$J_{j,T} = \int \rho_j(\mathbf{r}, t)\, \mu_T(dt, d^3r, d^3v). \tag{38}$$

Removing time labels can destroy the association with an evolving host; removing velocity can destroy information needed for a later dynamical kernel. Aggregated exposure therefore accompanies the trajectory and event record rather than replacing them.

D. Ensemble averages and incomplete histories

At fixed Q, nonnegative path integrands give the identities

$$\mathbb{E}[C_T \mid Q] = \int_0^T dt \int d^3r\, d^3v \times \mathbf{1}_{\{g(\mathbf{r},t)<0\}}\, p_{\mathrm{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q), \tag{39}$$

$$\mathbb{E}[J_{j,T} \mid Q] = \int_0^T dt \int d^3r\, d^3v \times \rho_j(\mathbf{r}, t)\, p_{\mathrm{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q). \tag{40}$$

If a density conditional on survival is used instead, it must be multiplied by $S(t \mid Q)$. Renormalizing the remaining trajectories silently changes the population being averaged. Equations (39) and (40) provide a future consistency comparison between trajectory accumulation and a transport-density solution.

The event record must distinguish no exit observed, entry or re-entry observed, finite-horizon censoring, declared escape, unresolved strong interactions, and loss of model validity. Escape can compete with return; it is not interchangeable with numerical failure. A spatial boundary crossing counts as escape only under the declared escape rule.

After an unresolved stop, complete-horizon residence and exposure remain unknown. The nonnegative accumulated integrals describe the modeled interval, while extrapolation needs a continuation model. Censoring can depend on the path and environment. Treating it as independent would be an additional statistical assumption requiring justification.

IV. CONDITIONING AND CALCULATION REQUIREMENTS

A. Physical and inferential uncertainty

Host uncertainty, launch uncertainty, and physical scattering are different sources of variation. For a trajectory observable O, their joint accounting takes the form

$$\mathbb{E}[O \mid \mathcal{D}, \mathcal{M}] = \int d\Theta\, p(\Theta \mid \mathcal{D}, \mathcal{M}) \times \int dz_0\, \pi_0(z_0 \mid \Theta, \mathcal{D}, \mathcal{M}) \times \mathbb{E}_{\mathrm{path}}[O \mid \Theta, z_0, \mathcal{D}, \mathcal{M}]. \tag{41}$$

The inner expectation includes the Wiener forcing and any additional randomness in a supplied stopping-event model. This factorization preserves conditional dependence. Merger selection can change host weights as well as launch properties. Alternative host families require separate conditioning or explicit model probabilities. Region definitions, reference densities, and any selection-dependent horizon must also be specified consistently.

Observed stellar kinematics constrain a host through an inference problem; they do not directly provide the entire potential, hidden remnant populations, or collision coefficients. Distribution-function models offer a possible route to consistency [4], but observational constraints, inferred functions, and simulation inputs must retain separate provenance. No posterior or model probabilities are evaluated here.

B. Information passed between the two steps

A calculation becomes defined only when its physical inputs and event conventions are supplied. These include the units and frame; species masses and distributions; internal and external gravity with boundary conditions; the joint launch law; collision closure and scale cutoffs; physical horizon; region geometry; and escape, strong-event, and validity criteria. An unspecified input is not a zero-valued effect.

Step 1 must preserve physical times, positions, velocities, remnant-mass labels, host and launch identifiers, original realization weights, and termination causes. A density-based calculation must preserve its conditioning and absorbed or killed probability. Reproducibility requires the source of each input and enough numerical information to reconstruct coefficient evaluation, stochastic realizations, and error control.

Step 2 attaches crossing events, residence, species exposures, and a time-augmented occupation measure. Every summary retains its denominator convention, modeled endpoint, and status. A path that becomes unresolved is not filled by an assumed central rest state. The same host and epoch must supply the density used for exposure and the distribution used to drive transport.

C. Acceptance criteria before physical interpretation

The following requirements specify validation to be performed; they are not reported as executed checks. Dimensional consistency, force–potential agreement, number-to-mass normalization, and positive-semidefinite diffusion must first be established. The implementation must preserve the Itô convention and account for probability lost at each boundary or stop. Appropriate conservative and analytically justified stochastic limits provide independent checks, including the energy identity in Eq. (24).

Entry and residence calculations must distinguish tangencies from crossings, handle interior, exterior, and boundary launches, and account for moving regions. Brief passages must remain resolved under a declared refinement procedure. Convergence is needed for the relevant distributions and accumulated observables, not solely for a few representative trajectories. Numerical and sampling errors must be separated from uncertainty in physical inputs.

Where both representations exist, path averages and the survival-weighted identities in Eqs. (39) and (40) should agree within declared numerical accuracy. Comparisons testing a physical simplification must hold the conditional population, horizon, region, and remaining inputs fixed. A useful result could identify a regime in which a reduction preserves the observables, or a discrepancy establishing which additional information is required.

Internal agreement does not calibrate the collision model. Comparison with independently resolved dynamics appropriate to the host and encounter regime is needed before interpreting predictions as quantitatively reliable. No solver, numerical tolerance, ensemble size, or passing threshold is assigned by the present formulation.

V. DISCUSSION

The central object of the formulation is a merger-conditioned, time-resolved distribution of remnant motion. Finite-region entry, residence, and exposure are functionals of that distribution. Keeping their definitions separate makes explicit why remaining bound, returning to a region, and spending time in a dense environment are different physical questions.

The description permits off-center launches, inherited progenitor motion, anisotropic and streaming stellar populations, time-dependent gravity, and stochastic velocity changes. These dependences are available in the equations; they are not all measured or independently adjustable parameters. A physical realization must justify how they are constrained together and where the adopted closure applies.

The formulation itself supplies no calibrated transport prediction. Its contribution is to organize the necessary assumptions, conditioning, and observables into one consistent initial-value and accumulation problem. Continued motion after entry, survival weighting, and termination status remain part of that problem rather than being hidden in a return-time prescription.

Timed position–velocity histories can eventually be combined with a separately justified companion population and interaction law. Residence and density exposure alone do not supply either ingredient. Companion acquisition, binary evolution, subsequent mergers, and inference from observed gravitational-wave sources therefore require additional physical descriptions beyond the two steps defined here.

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